Pearson Edexcel A-Level Mathematics (9MA0)

Module 3

Sequences & Trigonometry

Paper 1 & 2 (Pure) · 78 practice questions · answers and mark schemes at the back

Subtopic	Questions	Done
3.1 Sequences and Series	24	____ / 24
3.2 Radians & Trigonometry	21	____ / 21
3.3 Advanced Trigonometry & Modelling	18	____ / 18
3.4 Parametric Equations	15	____ / 15

How to use this workbook

1. Read the key facts and watch one video for the subtopic. Tick the skills checklist as you go.
2. Attempt each question in the squared working space without notes. Mark it with the scheme at the back.
3. Tick **Todo** when you have attempted it and **Mastered** only when you can redo it cold a week later.
4. Copy any mistakes into the notes box at the end of the subtopic. Rebuild the workbook with a new seed for fresh questions.

Name _____ Started _____ Built 14 Oct 2026, seed 2026

3.1 Sequences and Series

Arithmetic and geometric sequences and series, sum to infinity, sigma notation, recurrence relations and modelling with series.

Key facts

- AP: $u_n = a + (n-1)d$, $S_n = \frac{n}{2}(2a + (n-1)d) = \frac{n}{2}(a + l)$
- GP: $u_n = ar^{n-1}$, $S_n = \frac{a(1-r^n)}{1-r}$, $S_\infty = \frac{a}{1-r}$ for $|r| < 1$
- $\sum_{r=1}^{n} 1 = n$; a sum of a linear expression is an arithmetic series
- Recurrence $u_{n+1} = f(u_n)$: increasing if $u_{n+1} > u_n$ for all n; periodic if terms repeat

Method

- For 'find the first n such that...', form an inequality and use logs (GP) or a quadratic (AP).
- Count terms carefully: $\sum_{r=5}^{20}$ has 16 terms.
- In money problems, check whether the first year is $n = 1$ or $n = 0$.

Exam traps

- Using S_∞ when $|r| \geq 1$.
- Rounding n down when the question needs the first value exceeding a target.

In the exam: Paper 1 or 2, 6-10 marks; modelling contexts (salaries, savings, running distance) are popular.

Specification checklist

Skill	Learnt	Practised	Mastered
Arithmetic sequences	☐	☐	☐
Arithmetic series	☐	☐	☐
Geometric sequences	☐	☐	☐
Geometric series	☐	☐	☐
Sum to infinity	☐	☐	☐
Sigma notation	☐	☐	☐
Recurrence relations, increasing / decreasing / periodic	☐	☐	☐
Modelling with series	☐	☐	☐

Videos (click to open)

☐ **Recap** — Sequences & Series in 23 minutes • A-Level Maths, Pure Year 2, Chapter 3 (Bicen Maths)
☐ Arithmetic sequences — Arithmetic Sequences: Introducing Arithmetic Sequences (TLMaths)
☐ Arithmetic series — Arithmetic Sequences: Introducing Arithmetic Series (TLMaths)
☐ Geometric sequences — Geometric Sequences: Introducing Geometric Sequences (TLMaths)
☐ Geometric series — Geometric Sequences: Introducing Geometric Series (TLMaths)
☐ Sum to infinity — Geometric Sequences: Introducing Summing to Infinity (TLMaths)
☐ Sigma notation — Sigma Notation: Introducing Sigma Notation (TLMaths)
☐ Recurrence relations, increasing / decreasing / periodic — Sequences: Inductive Definitions and Recurrence Relations (TLMaths)
☐ Modelling with series — Modelling with Sequences: The Gardener Problem (TLMaths)

Q3.1.01 Geometric series ●○○ [2 marks] ☐ Todo ☐ Mastered

A geometric series has first term 29 and common ratio 1.2. Find the sum of the first 14 terms, giving your answer to 3 significant figures.

(2)

WORKING

Q3.1.02 Arithmetic sequences ●○○ [5 marks] ☐ Todo ☐ Mastered

The 8th term of an arithmetic sequence is 53 and the 21st term is 157.

(a) Find the first term and the common difference. **(3)**

(b) Find the sum of the first 30 terms. **(2)**

WORKING

Q3.1.03 Geometric sequences ●○○ [4 marks] ☐ Todo ☐ Mastered

The second term of a geometric sequence is 3 and the 5th term is $\frac{81}{64}$. Given that the common ratio is positive, find the common ratio and the first term.

(4)

WORKING

Q3.1.04 Geometric series ●○○ [2 marks] ☐ Todo ☐ Mastered

A geometric series has first term 7 and common ratio 0.8. Find the sum of the first 13 terms, giving your answer to 3 significant figures.

(2)

WORKING

Q3.1.05 Arithmetic sequences ●○○ [5 marks] ☐ Todo ☐ Mastered

The 5th term of an arithmetic sequence is 34 and the 13th term is 106.

(a) Find the first term and the common difference. **(3)**

(b) Find the sum of the first 47 terms. **(2)**

WORKING

Q3.1.06 Sigma notation ●○○ [3 marks] ☐ Todo ☐ Mastered

Find $\sum_{r=2}^{36}(6r+9)$.

(3)

WORKING

Q3.1.07 Geometric sequences ●○○ [4 marks] ☐ Todo ☐ Mastered

The second term of a geometric sequence is $\frac{87}{4}$ and the 5th term is $\frac{2349}{256}$. Given that the common ratio is positive, find the common ratio and the first term.

(4)

WORKING

Q3.1.08 **Recurrence relations, increasing / decreasing / periodic** ●○○ **[3 marks]** ☐ Todo ☐ Mastered

A sequence is defined by $u_1 = 2$, $u_{n+1} = -3u_n - 1$. Find u_2, u_3 and u_4, and state whether the sequence is increasing, decreasing or neither.

(3)

WORKING

Q3.1.09 **Geometric series** ●○○ **[2 marks]** ☐ Todo ☐ Mastered

A geometric series has first term 29 and common ratio 1.2. Find the sum of the first 8 terms, giving your answer to 3 significant figures.

(2)

WORKING

Q3.1.10 **Arithmetic sequences** ●○○ **[5 marks]** ☐ Todo ☐ Mastered

The 8th term of an arithmetic sequence is 57 and the 25th term is 176.

(a) Find the first term and the common difference. **(3)**

(b) Find the sum of the first 42 terms. **(2)**

WORKING

Q3.1.11 Geometric sequences ●○○ [4 marks] ☐ Todo ☐ Mastered

The second term of a geometric sequence is 78 and the 4th term is 702. Given that the common ratio is positive, find the common ratio and the first term.

(4)

WORKING

Q3.1.12 Modelling with series ●●○ [5 marks] ☐ Todo ☐ Mastered

Sam's salary in year 1 is £20,000. It increases by 4% each year.

(a) Find the first year in which Sam's salary exceeds £60,000. **(3)**

(b) Find the total Sam earns over the first 10 years, to the nearest £. **(2)**

WORKING

Q3.1.13 Recurrence relations, increasing / decreasing / periodic ●●○ [4 marks] ☐ Todo ☐ Mastered

A sequence is defined by $u_1 = 5$, $u_{n+1} = \frac{2}{u_n}$.

(a) Show that the sequence is periodic and state its order. **(2)**

(b) Find $\sum_{r=1}^{25} u_r$. **(2)**

WORKING

Q3.1.14 Sigma notation ●●○ [3 marks] ☐ Todo ☐ Mastered

Find the exact value of $\sum_{k=1}^{8} 8(3)^k$.

(3)

WORKING

Q3.1.15 Arithmetic series ●●○ [4 marks] ☐ Todo ☐ Mastered

An arithmetic series has first term 12 and common difference 7. The sum of the first n terms is 4335.
Show that $7n^2 + 17n - 8670 = 0$ and hence find n.

(4)

WORKING

Q3.1.16 Sum to infinity ●●○ [6 marks] ☐ Todo ☐ Mastered

A geometric series has second term 9 and sum to infinity 36.

(a) Show that $36r^2 - 36r + 9 = 0$. **(3)**

(b) Hence find the possible values of the common ratio, r, explaining why any value must be rejected. **(3)**

WORKING

Q3.1.17 **Modelling with series** ●●○ **[5 marks]** ☐ Todo ☐ Mastered

Sam's salary in year 1 is £26,000. It increases by 3% each year.

(a) Find the first year in which Sam's salary exceeds £40,000. **(3)**

(b) Find the total Sam earns over the first 10 years, to the nearest £. **(2)**

WORKING

Q3.1.18 **Recurrence relations, increasing / decreasing / periodic** ●●○ **[4 marks]** ☐ Todo ☐ Mastered

A sequence is defined by $u_1 = 3$, $u_{n+1} = \frac{2}{u_n}$.

(a) Show that the sequence is periodic and state its order. **(2)**

(b) Find $\sum_{r=1}^{41} u_r$. **(2)**

WORKING

Q3.1.19 Arithmetic series ●●○ [4 marks] ☐ Todo ☐ Mastered

An arithmetic series has first term 6 and common difference 4. The sum of the first n terms is 1680.
Show that $4n^2 + 8n - 3360 = 0$ and hence find n.

(4)

WORKING

Q3.1.20 Sum to infinity ●●○ [6 marks] ☐ Todo ☐ Mastered

A geometric series has second term -6 and sum to infinity 8.

(a) Show that $8r^2 - 8r - 6 = 0$. **(3)**

(b) Hence find the possible values of the common ratio, r, explaining why any value must be rejected. **(3)**

WORKING

Q3.1.21 Modelling with series ●●○ [5 marks] ☐ Todo ☐ Mastered

Sam's salary in year 1 is £26,000. It increases by 4% each year.

(a) Find the first year in which Sam's salary exceeds £45,000. **(3)**

(b) Find the total Sam earns over the first 10 years, to the nearest £. **(2)**

WORKING

Q3.1.22 Sigma notation ●●○ [3 marks] ☐ Todo ☐ Mastered

Find the exact value of $\sum_{k=1}^{11} 4\left(\frac{1}{2}\right)^k$.

(3)

WORKING

Q3.1.23 Arithmetic series ●●○ [4 marks] ☐ Todo ☐ Mastered

An arithmetic series has first term 10 and common difference 3. The sum of the first n terms is 413.
Show that $3n^2 + 17n - 826 = 0$ and hence find n.

(4)

WORKING

Q3.1.24 Sum to infinity ●●○ [6 marks] ☐ Todo ☐ Mastered

A geometric series has second term 4 and sum to infinity 18.

(a) Show that $18r^2 - 18r + 4 = 0$. **(3)**

(b) Hence find the possible values of the common ratio, r, explaining why any value must be rejected. **(3)**

WORKING

MY NOTES — 3.1 SEQUENCES AND SERIES: MISTAKES TO REMEMBER

3.2 Radians & Trigonometry

Radians, arcs and sectors, small angle approximations, sec/cosec/cot, Pythagorean identities and inverse trig functions.

Key facts

- π rad $= 180°$; arc $s = r\theta$, sector $A = \frac{1}{2}r^2\theta$, segment $= \frac{1}{2}r^2(\theta - \sin\theta)$
- Small θ (radians): $\sin\theta \approx \theta$, $\tan\theta \approx \theta$, $\cos\theta \approx 1 - \frac{\theta^2}{2}$
- $\sec x = \frac{1}{\cos x}$, $\operatorname{cosec} x = \frac{1}{\sin x}$, $\cot x = \frac{1}{\tan x}$
- $1 + \tan^2 x = \sec^2 x$, $1 + \cot^2 x = \operatorname{cosec}^2 x$
- arcsin: $[-1, 1] \to [-\frac{\pi}{2}, \frac{\pi}{2}]$; arccos: $[-1, 1] \to [0, \pi]$; arctan: $\mathbb{R} \to (-\frac{\pi}{2}, \frac{\pi}{2})$

Method

- Set calculators to radians whenever π or arc/sector formulas appear.
- Equations with $\sec^2$ and $\tan$: replace $\sec^2 x$ by $1 + \tan^2 x$ to get a quadratic in $\tan x$.
- Small angle: substitute, simplify, ignore powers of θ above the one asked for.

Exam traps

- Using degrees in $r\theta$.
- Missing solutions from the second quadrant (CAST).
- Small angle approximations only work in radians.

In the exam: Sector/segment questions are 5-7 marks; trig equations with identities appear on every paper.

Specification checklist

Skill	Learnt	Practised	Mastered
Radian measure	☐	☐	☐
Arc length	☐	☐	☐
Sector and segment areas	☐	☐	☐
Small angle approximations	☐	☐	☐
sec, cosec and cot	☐	☐	☐
Identities 1 + tan^2 = sec^2 and 1 + cot^2 = cosec^2	☐	☐	☐
Inverse trig functions arcsin, arccos, arctan	☐	☐	☐

Videos (click to open)

☐ **Recap** — Radians in less than 18 minutes • A-Level Maths, Pure Year 2, Chapter 5 (Bicen Maths)
☐ **Recap** — Trigonometry in 30 minutes • A-Level Maths, Pure Year 2, Chapters 6 & 7 (Bicen Maths)
☐ Radian measure — Trigonometry: Introducing Radians (TLMaths)
☐ Arc length — Trigonometry: Using the Formula for Arc Length in Radians (TLMaths)
☐ Sector and segment areas — Trigonometry: Using the Formula for Area of a Sector in Radians (TLMaths)
☐ Small angle approximations — Small-Angle Approximation: Geometrical Derivation (TLMaths)
☐ sec, cosec and cot — Trigonometry: Introducing cosec(x), sec(x) & cot(x) (TLMaths)
☐ Identities 1 + tan^2 = sec^2 and 1 + cot^2 = cosec^2 — Trigonometric Identities: Proving 1 + cot^2 θ = cosec^2 θ (TLMaths)
☐ Inverse trig functions arcsin, arccos, arctan — Trigonometry: Introducing arcsin(x) (TLMaths)

Q3.2.01 Radian measure ●○○ [2 marks] ☐ Todo ☐ Mastered

(a) Convert 150° to radians, giving your answer as an exact multiple of π. **(1)**

(b) Convert $\frac{5}{4}\pi$ radians to degrees. **(1)**

WORKING

Q3.2.02 sec, cosec and cot ●○○ [3 marks] ☐ Todo ☐ Mastered

Solve, for $0 \leq x < 360^\circ$, $\operatorname{cosec} x = -1.5$, giving your answers to 1 decimal place.

(3)

WORKING

Q3.2.03 Arc length ●○○ [4 marks] ☐ Todo ☐ Mastered

A sector OAB of a circle has radius 15 cm and angle 0.55 radians at the centre O.

(a) Find the length of the arc AB. **(1)**

(b) Find the perimeter of the sector. **(1)**

(c) Find the area of the sector. **(2)**

WORKING

Q3.2.04 Inverse trig functions arcsin, arccos, arctan ●○○ [3 marks] ☐ Todo ☐ Mastered

Find the exact value of $\arctan(-1)$ in radians, and state the domain and range of $y = \arctan x$.

(3)

WORKING

Q3.2.05 Radian measure ●○○ [2 marks] ☐ Todo ☐ Mastered

(a) Convert 315° to radians, giving your answer as an exact multiple of π. **(1)**

(b) Convert $\frac{3}{2}\pi$ radians to degrees. **(1)**

WORKING

Q3.2.06 sec, cosec and cot ●○○ [3 marks] ☐ Todo ☐ Mastered

Solve, for $0 \leq x < 360^\circ$, $\operatorname{cosec} x = 4$, giving your answers to 1 decimal place.

(3)

WORKING

Q3.2.07 Arc length ●○○ [4 marks] ☐ Todo ☐ Mastered

A sector OAB of a circle has radius 5 cm and angle 1.81 radians at the centre O.

(a) Find the length of the arc AB. **(1)**

(b) Find the perimeter of the sector. **(1)**

(c) Find the area of the sector. **(2)**

WORKING

Q3.2.08 Inverse trig functions arcsin, arccos, arctan ●○○ [3 marks] ☐ Todo ☐ Mastered

Find the exact value of $\arccos\left(\frac{\sqrt{3}}{2}\right)$ in radians, and state the domain and range of $y = \arccos x$.

(3)

WORKING

Q3.2.09 Radian measure ●○○ [2 marks] ☐ Todo ☐ Mastered

(a) Convert 270° to radians, giving your answer as an exact multiple of π. **(1)**

(b) Convert $\frac{2}{3}\pi$ radians to degrees. **(1)**

WORKING

Q3.2.10 sec, cosec and cot ●○○ [3 marks] ☐ Todo ☐ Mastered

Solve, for $0 \leq x < 360^\circ$, $\operatorname{cosec} x = 2$, giving your answers to 1 decimal place.

(3)

WORKING

Q3.2.11 Arc length ●○○ [4 marks] ☐ Todo ☐ Mastered

A sector OAB of a circle has radius 14 cm and angle 0.92 radians at the centre O.

(a) Find the length of the arc AB. **(1)**

(b) Find the perimeter of the sector. **(1)**

(c) Find the area of the sector. **(2)**

WORKING

Q3.2.12 Inverse trig functions arcsin, arccos, arctan ●○○ [3 marks] ☐ Todo ☐ Mastered

Find the exact value of $\arctan\left(\sqrt{3}\right)$ in radians, and state the domain and range of $y = \arctan x$.

(3)

WORKING

Q3.2.13 Identities 1 + tan^2 = sec^2 and 1 + cot^2 = cosec^2 ●●○ [5 marks] ☐ Todo ☐ Mastered

Solve, for $0 \leq x < 360^\circ$, $\sec^2 x - 2\tan x - 4 = 0$.

(5)

WORKING

Q3.2.14 Sector and segment areas ●●○ [4 marks] ☐ Todo ☐ Mastered

The chord AB subtends an angle of 2.28 radians at the centre O of a circle of radius 4 cm. Find the area of the minor segment cut off by AB, giving your answer to 3 significant figures.

(4)

WORKING

Q3.2.15 **Small angle approximations** ●●○ **[3 marks]** ☐ Todo ☐ Mastered

Given that θ is small, use the small angle approximations to find an approximation for $\frac{1-\cos 4\theta}{\theta \sin \theta}$ as a constant.

(3)

WORKING

Q3.2.16 **Identities 1 + tan^2 = sec^2 and 1 + cot^2 = cosec^2** ●●○ **[5 marks]** ☐ Todo ☐ Mastered

Solve, for $0 \leq x < 360^\circ$, $\sec^2 x - \frac{3}{2}\tan x - \frac{1}{2} = 0$.

(5)

WORKING

Q3.2.17 **Sector and segment areas** ●●○ **[4 marks]** ☐ Todo ☐ Mastered

The chord AB subtends an angle of 1.02 radians at the centre O of a circle of radius 5 cm. Find the area of the minor segment cut off by AB, giving your answer to 3 significant figures.

(4)

WORKING

Q3.2.18 Small angle approximations ●●○ [3 marks] ☐ Todo ☐ Mastered

Given that θ is small, use the small angle approximations to find an approximation for $\cos 5\theta + \theta \sin 2\theta$ in the form $p + q\theta^2$.

(3)

WORKING

Q3.2.19 Identities 1 + tan^2 = sec^2 and 1 + cot^2 = cosec^2 ●●○ [5 marks] ☐ Todo ☐ Mastered

Solve, for $0 \leq x < 360^\circ$, $\sec^2 x - 2 = 0$.

(5)

WORKING

Q3.2.20 Sector and segment areas ●●○ [4 marks] ☐ Todo ☐ Mastered

The chord AB subtends an angle of 1.19 radians at the centre O of a circle of radius 10 cm. Find the area of the minor segment cut off by AB, giving your answer to 3 significant figures.

(4)

WORKING

Q3.2.21 **Small angle approximations** ●●○ **[3 marks]** ☐ **Todo** ☐ **Mastered**

Given that θ is small, use the small angle approximations to find an approximation for $\cos 4\theta + \theta \sin 4\theta$ in the form $p + q\theta^2$.

(3)

WORKING

MY NOTES — 3.2 RADIANS & TRIGONOMETRY: MISTAKES TO REMEMBER

3.3 Advanced Trigonometry & Modelling

Addition and double angle formulae, the harmonic form $R\cos(x \pm \alpha)$, solving trig equations and proving identities, with modelling applications.

Key facts

- $\sin(A \pm B) = \sin A\cos B \pm \cos A\sin B$
- $\cos(A \pm B) = \cos A\cos B \mp \sin A\sin B$, $\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A\tan B}$
- $\sin 2A = 2\sin A\cos A$, $\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$
- $a\cos x + b\sin x = R\cos(x - \alpha)$ with $R = \sqrt{a^2 + b^2}$, $\tan\alpha = \frac{b}{a}$
- Maximum of $R\cos(x - \alpha)$ is R when $x = \alpha$; minimum $-R$

Method

- Solving $\sin(2x + 30^\circ) = k$: transform the interval first ($30 \le 2x + 30 < 750$), find all angles, then undo.
- Proofs: start from the more complicated side; convert everything to $\sin$ and $\cos$ if stuck.
- Harmonic form: expand $R\cos(x - \alpha)$, compare coefficients, divide to get $\tan\alpha$.

Exam traps

- Dividing by $\sin x$ or $\cos x$ and losing solutions (factorise instead).
- Giving α in degrees when the question works in radians (or vice versa).

In the exam: The R-form question is a Paper 2 regular (8-10 marks) often with a modelling part (tides, Ferris wheels, temperature).

Specification checklist

Skill	Learnt	Practised	Mastered
Addition formulae	☐	☐	☐
Double-angle formulae	☐	☐	☐
Solving trig equations	☐	☐	☐
Harmonic form R cos(x ± a), R sin(x ± a)	☐	☐	☐
Proving trig identities	☐	☐	☐
Modelling with trig functions	☐	☐	☐

Videos (click to open)

☐ **Recap** — Trigonometric Identities & Equations in less than 21 mins • A-Level Maths, Pure Year 1, Chapter 10 (Bicen Maths)
☐ **Recap** — Trigonometry in 30 minutes • A-Level Maths, Pure Year 2, Chapters 6 & 7 (Bicen Maths)
☐ Addition formulae — Compound Angles: Proving the Compound Angle Formulae (TLMaths)
☐ Double-angle formulae — Compound Angles: Introducing the Double Angle Formulae (TLMaths)
☐ Solving trig equations — Trig Equations: Solving Basic Trigonometric Equations in degrees (TLMaths)
☐ Harmonic form R cos(x ± a), R sin(x ± a) — Equivalent Forms: Writing acosθ + bsinθ in the form rcos(θ±α) or rsin(θ±α) (TLMaths)
☐ Proving trig identities — Proving Trigonometric Identities: The Methods (TLMaths)
☐ Modelling with trig functions — A level Edexcel June 2022 Paper 2 Q9 - Trigonometric functions - Modelling (ExamSolutions)

Q3.3.01 Addition formulae ●○○ [3 marks] ☐ Todo ☐ Mastered

Using an appropriate addition formula, find the exact value of $\cos 15^\circ$.

(3)

WORKING

Q3.3.02 Solving trig equations ●○○ [4 marks] ☐ Todo ☐ Mastered

Solve, for $0 \leq x < 180^\circ$, $\sin(2x) = 0.5$, giving your answers to 1 decimal place.

(4)

WORKING

Q3.3.03 Addition formulae ●○○ [3 marks] ☐ Todo ☐ Mastered

Using an appropriate addition formula, find the exact value of $\cos 105^\circ$.

(3)

WORKING

Q3.3.04 Proving trig identities ●●○ [4 marks] ☐ Todo ☐ Mastered

Prove that $\frac{\sin\theta}{1+\cos\theta} + \frac{1+\cos\theta}{\sin\theta} \equiv 2\operatorname{cosec}\theta$.

(4)

WORKING

Q3.3.05 Modelling with trig functions ●●○ [5 marks] ☐ Todo ☐ Mastered

The depth of water, D metres, in a harbour t hours after midnight is modelled by $D = 11 + 2.4\sin(30t)°$, $0 \leq t < 24$.

(a) State the maximum depth and find the first time it occurs. **(3)**

(b) A boat needs at least 11 m of water. For how long, in each 12-hour period, can it enter the harbour? **(2)**

WORKING

Q3.3.06 Double-angle formulae ●●○ [4 marks] ☐ Todo ☐ Mastered

Given that $\sin A = \frac{5}{13}$ and that A is acute, find the exact values of

(a) $\sin 2A$, (b) $\cos 2A$, (c) $\tan 2A$.

(a) **(2)**

(b) **(1)**

(c) **(1)**

WORKING

Q3.3.07 Solving trig equations ●●○ [6 marks] ☐ Todo ☐ Mastered

Solve, for $0 \leq x < 360°$, $3\sin^2 x - 2\cos x - 2 = 0$, giving your answers to 1 decimal place.

(6)

WORKING

Q3.3.08 Proving trig identities ●●○ [4 marks] ☐ Todo ☐ Mastered

Prove that $\cot\theta + \tan\theta \equiv 2\,\text{cosec}\,2\theta$.

(4)

WORKING

Q3.3.09 Modelling with trig functions ●●○ [5 marks] ☐ Todo ☐ Mastered

The depth of water, D metres, in a harbour t hours after midnight is modelled by $D = 8 + 4.1\sin(30t + 45)^\circ$, $0 \leq t < 24$.

(a) State the maximum depth and find the first time it occurs. **(3)**

(b) A boat needs at least 8 m of water. For how long, in each 12-hour period, can it enter the harbour? **(2)**

WORKING

Q3.3.10 Double-angle formulae ●●○ [4 marks] ☐ Todo ☐ Mastered

Given that $\sin A = \frac{7}{25}$ and that A is acute, find the exact values of
(a) $\sin 2A$, (b) $\cos 2A$, (c) $\tan 2A$.

(a) **(2)**

(b) **(1)**

(c) **(1)**

WORKING

Q3.3.11 Proving trig identities ●●○ [4 marks] ☐ Todo ☐ Mastered

Prove that $\frac{\cos 2\theta}{\cos\theta + \sin\theta} \equiv \cos\theta - \sin\theta$.

(4)

WORKING

Q3.3.12 Modelling with trig functions ●●○ [5 marks] ☐ Todo ☐ Mastered

The depth of water, D metres, in a harbour t hours after midnight is modelled by $D = 9 + 4.4\sin(30t)^\circ$, $0 \leq t < 24$.

(a) State the maximum depth and find the first time it occurs. **(3)**

(b) A boat needs at least 9 m of water. For how long, in each 12-hour period, can it enter the harbour? **(2)**

WORKING

Q3.3.13 Double-angle formulae ●●○ [4 marks] ☐ Todo ☐ Mastered

Given that $\sin A = \frac{20}{29}$ and that A is acute, find the exact values of
(a) $\sin 2A$, (b) $\cos 2A$, (c) $\tan 2A$.

(a) **(2)**

(b) **(1)**

(c) **(1)**

WORKING

Q3.3.14 Proving trig identities ●●○ [4 marks] ☐ Todo ☐ Mastered

Prove that $\sin 3\theta \equiv 3\sin\theta - 4\sin^3\theta$.

(4)

WORKING

Q3.3.15 Modelling with trig functions ●●○ [5 marks] ☐ Todo ☐ Mastered

The depth of water, D metres, in a harbour t hours after midnight is modelled by $D = 8 + 3.1\sin(30t + 30)^\circ$, $0 \leq t < 24$.

(a) State the maximum depth and find the first time it occurs. **(3)**

(b) A boat needs at least 8 m of water. For how long, in each 12-hour period, can it enter the harbour? **(2)**

WORKING

Q3.3.16 Harmonic form R cos(x ± a), R sin(x ± a) ●●● [8 marks] ☐ Todo ☐ Mastered

(a) Express $12\cos x - 5\sin x$ in the form $R\cos(x+\alpha)$, where $R > 0$ and $0 < \alpha < 90^\circ$. Give α to 2 decimal places. **(3)**

(b) Hence solve $12\cos x - 5\sin x = 6.4$ for $0 \leq x < 360^\circ$, giving answers to 1 decimal place. **(4)**

(c) State the maximum value of $12\cos x - 5\sin x$. **(1)**

WORKING

Q3.3.17 Harmonic form R cos(x ± a), R sin(x ± a) ●●● [8 marks] ☐ Todo ☐ Mastered

(a) Express $3\cos x + 4\sin x$ in the form $R\cos(x-\alpha)$, where $R > 0$ and $0 < \alpha < 90^\circ$. Give α to 2 decimal places. **(3)**

(b) Hence solve $3\cos x + 4\sin x = 2.0$ for $0 \leq x < 360^\circ$, giving answers to 1 decimal place. **(4)**

(c) State the maximum value of $3\cos x + 4\sin x$. **(1)**

WORKING

Q3.3.18 Harmonic form R cos(x ± a), R sin(x ± a) ●●● [8 marks] ☐ Todo ☐ Mastered

(a) Express $2\sin x - 3\cos x$ in the form $R\sin(x - \alpha)$, where $R > 0$ and $0 < \alpha < 90^\circ$. Give α to 2 decimal places. **(3)**

(b) Hence solve $2\sin x - 3\cos x = 2.7$ for $0 \leq x < 360^\circ$, giving answers to 1 decimal place. **(4)**

(c) State the maximum value of $2\sin x - 3\cos x$. **(1)**

WORKING

MY NOTES — 3.3 ADVANCED TRIGONOMETRY & MODELLING: MISTAKES TO REMEMBER

3.4 Parametric Equations

Parametric equations: converting to Cartesian form, sketching, finding intersections, and modelling motion with a parameter.

Key facts

- Eliminate t: make t the subject of one equation and substitute
- Trig parameters: use $\sin^2 t + \cos^2 t = 1$, $\sec^2 t = 1 + \tan^2 t$, $\cos 2t = 1 - 2\sin^2 t$
- $x = a\cos t + p$, $y = a\sin t + q$ is the circle $(x-p)^2 + (y-q)^2 = a^2$
- The domain of the Cartesian equation comes from the range of $x(t)$

Method

- For intersections with a line, substitute $x(t)$ and $y(t)$ into the line and solve for t.
- For axis crossings, set $x(t) = 0$ or $y(t) = 0$.
- In modelling, interpret t (often time) and check the realistic range of t.

Exam traps

- Forgetting to state the domain/range restrictions.
- Only giving t values when coordinates are asked for.

In the exam: Paper 2, 6-10 marks; often joined with parametric differentiation or integration.

Specification checklist

Skill	Learnt	Practised	Mastered
Parametric to Cartesian	☐	☐	☐
Trig parametric conversions	☐	☐	☐
Sketching parametric curves	☐	☐	☐
Intersections with parametric curves	☐	☐	☐
Modelling with parametric equations	☐	☐	☐

Videos (click to open)

☐ **Recap** — Parametric Equations in less than 21 minutes • A-Level Maths, Pure Year 2, Chapter 8 (Bicen Maths)
☐ Parametric to Cartesian — Parametric Equations: Parametric to Cartesian (TLMaths)
☐ Sketching parametric curves — Parametric Equations: Introducing Parametric Equations (TLMaths)
☐ Intersections with parametric curves — Finding points of Intersection between Parametric and Cartesian Equations (ExamSolutions)
☐ Modelling with parametric equations — Parametric Equation Modelling: A Projectiles Problem (TLMaths)

Q3.4.01 Parametric to Cartesian ●○○ [2 marks] ☐ Todo ☐ Mastered

A curve has parametric equations $x = 3t + 1$, $y = t^2 + 3$. Find a Cartesian equation of the curve in the form $y = f(x)$.

(2)

WORKING

Q3.4.02 Sketching parametric curves ●○○ [3 marks] ☐ Todo ☐ Mastered

A curve has parametric equations $x = 3t + 4$, $y = t^2 - 25$. Find the coordinates of the points where the curve meets the x-axis, and the point where $t = 0$.

(3)

WORKING

Q3.4.03 Parametric to Cartesian ●○○ [2 marks] ☐ Todo ☐ Mastered

A curve has parametric equations $x = t - 2$, $y = t^2 - 3$. Find a Cartesian equation of the curve in the form $y = f(x)$.

(2)

WORKING

Q3.4.04 Sketching parametric curves ●○○ [3 marks] ☐ Todo ☐ Mastered

A curve has parametric equations $x = 3 - 3t$, $y = t^2 - 25$. Find the coordinates of the points where the curve meets the x-axis, and the point where $t = 0$.

(3)

WORKING

Q3.4.05 Parametric to Cartesian ●○○ [2 marks] ☐ Todo ☐ Mastered

A curve has parametric equations $x = 4 - t$, $y = t^2 - 1$. Find a Cartesian equation of the curve in the form $y = f(x)$.

(2)

WORKING

Q3.4.06 Sketching parametric curves ●○○ [3 marks] ☐ Todo ☐ Mastered

A curve has parametric equations $x = 2t + 1$, $y = t^2 - 4$. Find the coordinates of the points where the curve meets the x-axis, and the point where $t = 0$.

(3)

WORKING

Q3.4.07 Intersections with parametric curves ●●○ [4 marks] ☐ Todo ☐ Mastered

The curve C has parametric equations $x = t^2$, $y = 2t$. The line $y = 1x - 8$ meets C at two points. Find the coordinates of these points.

(4)

WORKING

Q3.4.08 Modelling with parametric equations ●●○ [5 marks] ☐ Todo ☐ Mastered

A ball is kicked from ground level. Its position after t seconds is modelled by $x = 12.30t$, $y = 18.24t - 4.9t^2$ (metres).

(a) Find the time when the ball lands. **(2)**

(b) Find the horizontal distance travelled. **(1)**

(c) Find a Cartesian equation of its path. **(2)**

WORKING

Q3.4.09 Trig parametric conversions ●●○ [3 marks] ☐ Todo ☐ Mastered

A curve has parametric equations $x = \sin t$, $y = 4\cos 2t$, $-\frac{\pi}{2} \leq t \leq \frac{\pi}{2}$. Show that a Cartesian equation is $y = 4 - 8x^2$ and state the domain of x.

(3)

WORKING

Q3.4.10 Intersections with parametric curves ●●○ [4 marks] ☐ Todo ☐ Mastered

The curve C has parametric equations $x = t^2$, $y = 2t$. The line $y = -\frac{1}{3}x - \frac{8}{3}$ meets C at two points. Find the coordinates of these points.

(4)

WORKING

Q3.4.11 Modelling with parametric equations ●●○ [5 marks] ☐ Todo ☐ Mastered

A ball is kicked from ground level. Its position after t seconds is modelled by $x = 14.38t$, $y = 10.83t - 4.9t^2$ (metres).

(a) Find the time when the ball lands. **(2)**

(b) Find the horizontal distance travelled. **(1)**

(c) Find a Cartesian equation of its path. **(2)**

WORKING

Q3.4.12 Trig parametric conversions ●●○ [3 marks] ☐ Todo ☐ Mastered

A curve has parametric equations $x = 6\sec t$, $y = 5\tan t$. Find a Cartesian equation of the curve.

(3)

WORKING

Q3.4.13 Intersections with parametric curves ●●○ [4 marks] ☐ Todo ☐ Mastered

The curve C has parametric equations $x = t^2$, $y = 2t$. The line $y = -\frac{1}{2}x$ meets C at two points. Find the coordinates of these points.

(4)

WORKING

Q3.4.14 Modelling with parametric equations ●●○ [5 marks] ☐ Todo ☐ Mastered

A ball is kicked from ground level. Its position after t seconds is modelled by $x = 5.00t$, $y = 8.66t - 4.9t^2$ (metres).

(a) Find the time when the ball lands. **(2)**

(b) Find the horizontal distance travelled. **(1)**

(c) Find a Cartesian equation of its path. **(2)**

WORKING

Q3.4.15 Trig parametric conversions ●●○ [3 marks] ☐ Todo ☐ Mastered

A curve has parametric equations $x = 4\sec t$, $y = 3\tan t$. Find a Cartesian equation of the curve.

(3)

WORKING

MY NOTES — 3.4 PARAMETRIC EQUATIONS: MISTAKES TO REMEMBER

Answers and mark schemes

M = method mark, A = accuracy mark (depends on the preceding M), B = independent mark, ft = follow-through, * = given answer. Numerical answers are to 3 s.f. unless stated.

3.1 Sequences and Series

Q3.1.01 Geometric series · 2 marks

1720

M1 $\frac{29(1-1.2^{14})}{1-1.2}$ · A1

Q3.1.02 Arithmetic sequences · 5 marks

(a) $a=-3,\ d=8$

M1 $a+7d=53$ and $a+20d=157$ · A1 d · A1 a

(b) $S_{30}=3390$

M1 $\frac{30}{2}(2a+29d)$ · A1

Q3.1.03 Geometric sequences · 4 marks

$r=\frac{3}{4},\ a=4$

M1 $\frac{ar^4}{ar}=r^3$ · A1 $r=\frac{3}{4}$ · M1 substitutes back · A1 a

Q3.1.04 Geometric series · 2 marks

33.1

M1 $\frac{7(1-0.8^{13})}{1-0.8}$ · A1

Q3.1.05 Arithmetic sequences · 5 marks

(a) $a=-2,\ d=9$

M1 $a+4d=34$ and $a+12d=106$ · A1 d · A1 a

(b) $S_{47}=9635$

M1 $\frac{47}{2}(2a+46d)$ · A1

Q3.1.06 Sigma notation · 3 marks

4305

M1 recognises an arithmetic series with 35 terms · M1 first term 21, last term 225 · A1

Q3.1.07 Geometric sequences · 4 marks

$r=\frac{3}{4},\ a=29$

M1 $\frac{ar^4}{ar}=r^3$ · A1 $r=\frac{3}{4}$ · M1 substitutes back · A1 a

Q3.1.08 Recurrence relations, increasing / decreasing / periodic · 3 marks

$u_2=-7,\ u_3=20,\ u_4=-61$; neither

M1 uses the recurrence · A1 all three terms · B1 description

Q3.1.09 Geometric series · 2 marks

478

M1 $\frac{29(1-1.2^{8})}{1-1.2}$ · A1

Q3.1.10 Arithmetic sequences · 5 marks

(a) $a=8,\ d=7$

M1 $a+7d=57$ and $a+24d=176$ · A1 d · A1 a

(b) $S_{42}=6363$

M1 $\frac{42}{2}(2a+41d)$ · A1

Q3.1.11 Geometric sequences · 4 marks

$r=3,\ a=26$

M1 $\frac{ar^3}{ar}=r^2$ · A1 $r=3$ · M1 substitutes back · A1 a

Q3.1.12 Modelling with series · 5 marks

(a) year 30

M1 $20000(1.04)^{n-1}>60000$ · M1 logs · A1 year 30

(b) £240,122

M1 geometric sum · A1

Q3.1.13 Recurrence relations, increasing / decreasing / periodic · 4 marks

(a) $u_2=\frac{2}{5},\ u_3=5$; period 2

M1 finds $u_2,\ u_3$ · A1 order 2

(b) $\frac{349}{5}$

M1 counts complete cycles · A1

Q3.1.14 Sigma notation · 3 marks

78720

M1 geometric with $a=24$, $r=3$ · M1 sum formula · A1

Q3.1.15 Arithmetic series · 4 marks

$n=34$

M1 uses $S_n=\frac{n}{2}(2a+(n-1)d)$ · A1* rearranges to the given quadratic · M1 solves · A1 $n=34$ (reject negative)

Q3.1.16 Sum to infinity · 6 marks

(a) given

M1 $ar=9$ · M1 $\frac{a}{1-r}=36$ · A1* eliminates a

(b) $r=\frac{1}{2}$

M1 solves the quadratic · A1 roots · B1 requires $|r|<1$ for a sum to infinity

Q3.1.17 Modelling with series · 5 marks

(a) year 16

M1 $26000(1.03)^{n-1}>40000$ · M1 logs · A1 year 16

(b) £298,061

M1 geometric sum · A1

Q3.1.18 Recurrence relations, increasing / decreasing / periodic · 4 marks

(a) $u_2=\frac{2}{3}$, $u_3=3$; period 2

M1 finds u_2, u_3 · A1 order 2

(b) $\frac{229}{3}$

M1 counts complete cycles · A1

Q3.1.19 Arithmetic series · 4 marks

$n=28$

M1 uses $S_n=\frac{n}{2}(2a+(n-1)d)$ · A1* rearranges to the given quadratic · M1 solves · A1 $n=28$ (reject negative)

Q3.1.20 Sum to infinity · 6 marks

(a) given

M1 $ar=-6$ · M1 $\frac{a}{1-r}=8$ · A1* eliminates a

(b) $r=-\frac{1}{2}$ (reject $r=\frac{3}{2}$ since $|r|\geq 1$)

M1 solves the quadratic · A1 roots · B1 requires $|r|<1$ for a sum to infinity

Q3.1.21 Modelling with series · 5 marks

(a) year 15

M1 $26000(1.04)^{n-1}>45000$ · M1 logs · A1 year 15

(b) £312,159

M1 geometric sum · A1

Q3.1.22 Sigma notation · 3 marks

$\frac{2047}{512}$

M1 geometric with $a=2$, $r=\frac{1}{2}$ · M1 sum formula · A1

Q3.1.23 Arithmetic series · 4 marks

$n=14$

M1 uses $S_n=\frac{n}{2}(2a+(n-1)d)$ · A1* rearranges to the given quadratic · M1 solves · A1 $n=14$ (reject negative)

Q3.1.24 Sum to infinity · 6 marks

(a) given

M1 $ar=4$ · M1 $\frac{a}{1-r}=18$ · A1* eliminates a

(b) $r=\frac{1}{3}, \frac{2}{3}$

M1 solves the quadratic · A1 roots · B1 requires $|r|<1$ for a sum to infinity

3.2 Radians & Trigonometry

Q3.2.01 Radian measure · 2 marks

(a) $\frac{5}{6}\pi$

B1

(b) 225°

B1

Q3.2.02 sec, cosec and cot · 3 marks

$x=180.0^\circ, 221.8^\circ, 318.2^\circ$

M1 $\sin x=-\frac{2}{3}$ · A1 one value · A1 all values

Q3.2.03 **Arc length · 4 marks**

(a) 8.25 cm

B1 $r\theta$

(b) 38.2 cm

B1ft

(c) 61.9 cm^2

M1 $\frac{1}{2}r^2\theta$ · A1

Q3.2.04 **Inverse trig functions arcsin, arccos, arctan · 3 marks**

$-\frac{\pi}{4}$; domain $x \in \mathbb{R}$, range $-\frac{\pi}{2} < y < \frac{\pi}{2}$

B1 value · B1 domain · B1 range

Q3.2.05 **Radian measure · 2 marks**

(a) $\frac{7}{4}\pi$

B1

(b) 270°

B1

Q3.2.06 **sec, cosec and cot · 3 marks**

$x = 14.5^\circ, 165.5^\circ, 180.0^\circ$

M1 $\sin x = \frac{1}{4}$ · A1 one value · A1 all values

Q3.2.07 **Arc length · 4 marks**

(a) 9.05 cm

B1 $r\theta$

(b) 19.1 cm

B1ft

(c) 22.6 cm^2

M1 $\frac{1}{2}r^2\theta$ · A1

Q3.2.08 **Inverse trig functions arcsin, arccos, arctan · 3 marks**

$\frac{\pi}{6}$; domain $-1 \leq x \leq 1$, range $0 \leq y \leq \pi$

B1 value · B1 domain · B1 range

Q3.2.09 **Radian measure · 2 marks**

(a) $\frac{3}{2}\pi$

B1

(b) 120°

B1

Q3.2.10 **sec, cosec and cot · 3 marks**

$x = 30.0^\circ, 150.0^\circ, 180.0^\circ$

M1 $\sin x = \frac{1}{2}$ · A1 one value · A1 all values

Q3.2.11 **Arc length · 4 marks**

(a) 12.9 cm

B1 $r\theta$

(b) 40.9 cm

B1ft

(c) 90.2 cm^2

M1 $\frac{1}{2}r^2\theta$ · A1

Q3.2.12 **Inverse trig functions arcsin, arccos, arctan · 3 marks**

$\frac{\pi}{3}$; domain $x \in \mathbb{R}$, range $-\frac{\pi}{2} < y < \frac{\pi}{2}$

B1 value · B1 domain · B1 range

Q3.2.13 **Identities 1 + tan^2 = sec^2 and 1 + cot^2 = cosec^2 · 5 marks**

$x = 71.6^\circ, 135.0^\circ, 251.6^\circ, 315.0^\circ$

M1 uses $\sec^2 x = 1 + \tan^2 x$ · M1 $\tan^2 x - 2\tan x - 3 = 0$ · A1 $\tan x = -1, 3$ · M1 finds angles in range · A1 all

Q3.2.14 **Sector and segment areas · 4 marks**

12.2 cm^2

M1 sector area $\frac{1}{2}r^2\theta$ · M1 triangle area $\frac{1}{2}r^2\sin\theta$ · M1 subtracts · A1

Q3.2.15 **Small angle approximations · 3 marks**

8

M1 uses $\sin\theta \approx \theta$, $\cos\theta \approx 1 - \frac{\theta^2}{2}$, $\tan\theta \approx \theta$ · M1 simplifies · A1

Q3.2.16 **Identities 1 + tan^2 = sec^2 and 1 + cot^2 = cosec^2 · 5 marks**

$x = 26.6^\circ, 45.0^\circ, 206.6^\circ, 225.0^\circ$

M1 uses $\sec^2 x = 1 + \tan^2 x$ · M1 $\tan^2 x - \frac{3}{2}\tan x + \frac{1}{2} = 0$ · A1 $\tan x = \frac{1}{2}, 1$ · M1 finds angles in range · A1 all

Q3.2.17 **Sector and segment areas · 4 marks**

$2.10\ \text{cm}^2$

M1 sector area $\frac{1}{2}r^2\theta$ · M1 triangle area $\frac{1}{2}r^2\sin\theta$ · M1 subtracts · A1

Q3.2.18 **Small angle approximations · 3 marks**

$1-\frac{21\theta^2}{2}$

M1 uses $\sin\theta\approx\theta$, $\cos\theta\approx1-\frac{\theta^2}{2}$, $\tan\theta\approx\theta$ · M1 simplifies · A1

Q3.2.19 **Identities 1 + tan^2 = sec^2 and 1 + cot^2 = cosec^2 · 5 marks**

$x=45.0^\circ, 135.0^\circ, 225.0^\circ, 315.0^\circ$

M1 uses $\sec^2 x=1+\tan^2 x$ · M1 $\tan^2 x-1=0$ · A1 $\tan x=1,\ -1$ · M1 finds angles in range · A1 all

Q3.2.20 **Sector and segment areas · 4 marks**

$13.1\ \text{cm}^2$

M1 sector area $\frac{1}{2}r^2\theta$ · M1 triangle area $\frac{1}{2}r^2\sin\theta$ · M1 subtracts · A1

Q3.2.21 **Small angle approximations · 3 marks**

$1-4\theta^2$

M1 uses $\sin\theta\approx\theta$, $\cos\theta\approx1-\frac{\theta^2}{2}$, $\tan\theta\approx\theta$ · M1 simplifies · A1

3.3 Advanced Trigonometry & Modelling

Q3.3.01 **Addition formulae · 3 marks**

$\frac{\sqrt{6}+\sqrt{2}}{4}$

M1 writes $15=45-30$ · M1 correct expansion with exact values · A1

Q3.3.02 **Solving trig equations · 4 marks**

$x=15.0^\circ, 75.0^\circ$

M1 adjusts the interval for $2x$ · A1 one correct angle · M1 uses symmetry/period to find others · A1 all, no extras

Q3.3.03 **Addition formulae · 3 marks**

$\frac{\sqrt{2}-\sqrt{6}}{4}$

M1 writes $105=60+45$ · M1 correct expansion with exact values · A1

Q3.3.04 **Proving trig identities · 4 marks**

Proof using common denominator, $\sin^2+\cos^2=1$

M1 correct identity used · M1 manipulates towards RHS · M1 second identity/simplification · A1* fully correct proof

Q3.3.05 **Modelling with trig functions · 5 marks**

(a) 13.4 m at $t=3$ hours

B1 max · M1 sets the angle to 90° · A1 time

(b) 6 hours

M1 $\sin\geq0$ for half of each cycle · A1

Q3.3.06 **Double-angle formulae · 4 marks**

(a) $\frac{120}{169}$

M1 $\cos A=\frac{12}{13}$ · A1

(b) $\frac{119}{169}$

B1

(c) $\frac{120}{119}$

B1ft

Q3.3.07 **Solving trig equations · 6 marks**

$x=70.5^\circ, 180.0^\circ, 289.5^\circ$

M1 uses $\sin^2 x=1-\cos^2 x$ · A1 $3\cos^2 x+2\cos x-1=0$ · M1 solves quadratic · A1 $\cos x=\frac{1}{3},\ -1$ · M1 finds angles · A1 all, no extras

Q3.3.08 **Proving trig identities · 4 marks**

Proof using write in sin, cos; use $\sin 2\theta$

M1 correct identity used · M1 manipulates towards RHS · M1 second identity/simplification · A1* fully correct proof

Q3.3.09 **Modelling with trig functions · 5 marks**

(a) 12.1 m at $t=\frac{3}{2}$ hours

B1 max · M1 sets the angle to 90° · A1 time

(b) 6 hours

M1 $\sin\geq0$ for half of each cycle · A1

Q3.3.10 Double-angle formulae · 4 marks
(a) $\frac{336}{625}$
M1 $\cos A = \frac{24}{25}$ · A1
(b) $\frac{527}{625}$
B1
(c) $\frac{336}{527}$
B1ft

Q3.3.11 Proving trig identities · 4 marks
Proof using $\cos 2\theta = \cos^2 - \sin^2$, difference of two squares
M1 correct identity used · M1 manipulates towards RHS · M1 second identity/simplification · A1* fully correct proof

Q3.3.12 Modelling with trig functions · 5 marks
(a) 13.4 m at $t = 3$ hours
B1 max · M1 sets the angle to $90°$ · A1 time
(b) 6 hours
M1 $\sin \geq 0$ for half of each cycle · A1

Q3.3.13 Double-angle formulae · 4 marks
(a) $\frac{840}{841}$
M1 $\cos A = \frac{21}{29}$ · A1
(b) $\frac{41}{841}$
B1
(c) $\frac{840}{41}$
B1ft

Q3.3.14 Proving trig identities · 4 marks
Proof using $\sin(2\theta + \theta)$, then double-angle formulae
M1 correct identity used · M1 manipulates towards RHS · M1 second identity/simplification · A1* fully correct proof

Q3.3.15 Modelling with trig functions · 5 marks
(a) 11.1 m at $t = 2$ hours
B1 max · M1 sets the angle to $90°$ · A1 time
(b) 6 hours
M1 $\sin \geq 0$ for half of each cycle · A1

Q3.3.16 Harmonic form R cos(x ± a), R sin(x ± a) · 8 marks
(a) $R = 13$, $\alpha = 22.62°$
B1 R · M1 $\tan\alpha = \frac{5}{12}$ · A1 α
(b) $x = 37.9°, 276.9°$
M1 divides by R and uses inverse function · A1 first angle · M1 second angle in range · A1 both
(c) 13
B1ft

Q3.3.17 Harmonic form R cos(x ± a), R sin(x ± a) · 8 marks
(a) $R = 5$, $\alpha = 53.13°$
B1 R · M1 $\tan\alpha = \frac{4}{3}$ · A1 α
(b) $x = 119.6°, 346.7°$
M1 divides by R and uses inverse function · A1 first angle · M1 second angle in range · A1 both
(c) 5
B1ft

Q3.3.18 Harmonic form R cos(x ± a), R sin(x ± a) · 8 marks
(a) $R = \sqrt{13}$, $\alpha = 56.31°$
B1 R · M1 $\tan\alpha = \frac{3}{2}$ · A1 α
(b) $x = 104.8°, 187.8°$
M1 divides by R and uses inverse function · A1 first angle · M1 second angle in range · A1 both
(c) $\sqrt{13}$
B1ft

3.4 Parametric Equations

Q3.4.01 Parametric to Cartesian · 2 marks
$y = \frac{x^2}{9} - \frac{2x}{9} + \frac{28}{9}$
M1 $t = \frac{x-1}{3}$ · A1

Q3.4.02 Sketching parametric curves · 3 marks
$(-11, 0)$, $(19, 0)$; $t = 0$ gives $(4, -25)$
M1 $y = 0 \Rightarrow t = \pm 5$ · A1 both points · B1

Q3.4.03 **Parametric to Cartesian · 2 marks**

$y = x^2 + 4x + 1$

M1 $t = \frac{x+2}{1}$ · A1

Q3.4.04 **Sketching parametric curves · 3 marks**

$(-12, 0)$, $(18, 0)$; $t = 0$ gives $(3, -25)$

M1 $y = 0 \Rightarrow t = \pm 5$ · A1 both points · B1

Q3.4.05 **Parametric to Cartesian · 2 marks**

$y = x^2 - 8x + 15$

M1 $t = \frac{x-4}{-1}$ · A1

Q3.4.06 **Sketching parametric curves · 3 marks**

$(-3, 0)$, $(5, 0)$; $t = 0$ gives $(1, -4)$

M1 $y = 0 \Rightarrow t = \pm 2$ · A1 both points · B1

Q3.4.07 **Intersections with parametric curves · 4 marks**

$(4, -4)$ and $(16, 8)$

M1 substitutes parametric forms into the line · A1 quadratic in t · M1 solves · A1 both points

Q3.4.08 **Modelling with parametric equations · 5 marks**

(a) $t = 3.72$ s

M1 $y = 0$ · A1

(b) 45.8 m

B1ft

(c) $y = 1.48x - 0.0324x^2$

M1 eliminates t · A1

Q3.4.09 **Trig parametric conversions · 3 marks**

$y = 4(1 - 2\sin^2 t) = 4 - 8x^2, -1 \leq x \leq 1$

M1 $\cos 2t = 1 - 2\sin^2 t$ · A1* · B1 domain

Q3.4.10 **Intersections with parametric curves · 4 marks**

$(16, -8)$ and $(4, -4)$

M1 substitutes parametric forms into the line · A1 quadratic in t · M1 solves · A1 both points

Q3.4.11 **Modelling with parametric equations · 5 marks**

(a) $t = 2.21$ s

M1 $y = 0$ · A1

(b) 31.8 m

B1ft

(c) $y = 0.754x - 0.0237x^2$

M1 eliminates t · A1

Q3.4.12 **Trig parametric conversions · 3 marks**

$\frac{x^2}{36} - \frac{y^2}{25} = 1$

M1 $\sec^2 t - \tan^2 t = 1$ · M1 substitutes · A1

Q3.4.13 **Intersections with parametric curves · 4 marks**

$(16, -8)$ and $(0, 0)$

M1 substitutes parametric forms into the line · A1 quadratic in t · M1 solves · A1 both points

Q3.4.14 **Modelling with parametric equations · 5 marks**

(a) $t = 1.77$ s

M1 $y = 0$ · A1

(b) 8.84 m

B1ft

(c) $y = 1.73x - 0.196x^2$

M1 eliminates t · A1

Q3.4.15 **Trig parametric conversions · 3 marks**

$\frac{x^2}{16} - \frac{y^2}{9} = 1$

M1 $\sec^2 t - \tan^2 t = 1$ · M1 substitutes · A1